MEMORANDUM
OFFICE OF THE STATE ENGINEER
Hydrology Bureau

DATE: **May 24, 2023**

TO: **Ramona Martinez, District VI Manager, Water Rights Division**
Lorraine Garcia, Upper Rio Grande Basin Manager, Water Rights Division
Chris Stageman, Interstate Stream Commission
Cindy Stokes, Interstate Stream Commission

FROM: **Laura H. Petronis, Hydrology Bureau** LHP

THROUGH: **Katie Zemlick, Ph.D., Hydrology Bureau Chief** KZ

SUBJECT: **Calendar Year 2022 hydrologic impacts on the Rio Grande, Rio Pojoaque-Nambe, Rio Tesuque, and La Cienega springs due to Buckman well field pumping, RG-20516 et al.**

This memorandum describes the estimated effects on the Rio Grande, Rio Pojoaque-Nambe and Rio Tesuque due to calendar year (CY) 1972 through 2022 Buckman well field pumping and the estimated effects on La Cienega springs due to CY1988 through CY2022 Buckman well field pumping.

CY1988 to CY2022 diversions from the Buckman well field reported by the City of Santa Fe are presented in Table 1. Monthly CY2022 Buckman diversions reported by the City of Santa Fe are presented in Table 2. CY2022 pumping totaled 975.47 acre-feet (ac-ft), which represents the ninth lowest annual pumping from the Buckman well field since 1988 (the earliest year considered in Table 1). Newer wells 10 through 13 supplied 262.20 ac-ft (26.9 percent) of the 2022 diversion from the Buckman well field. The three wells (Buckman 1, 7, and 8) in the model cell closest to the Rio Grande (row, column: 13, 11) supplied a total of 654.95 ac-ft (67.1 percent) of the 2022 Buckman well field diversion.

Simulated stream depletions on the Rio Grande and Santa Fe-area tributaries

Streamflow depletion on the Rio Grande, Rio Pojoaque-Nambe and Rio Tesuque due to Buckman well field pumping from CY1972 to CY2021 was estimated by following the general procedure described in Compton (1988). It specifies that residual hydrologic effects on the Rio Grande and its tributaries resulting from CY1972 through CY1987 Buckman well field pumping be calculated using the NMOSE analytical model,[1] and that post-1987 pumping effects on nearby streams be calculated using the McAda and Wasiolek (1988) ground-water flow model.

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[1] Residual hydrologic effects on the Rio Pojoaque-Nambe and Rio Tesuque due to CY1972-CY1987 Buckman well field pumping were calculated for CY1988 through CY2010 using the NMOSE analytical model and described in a letter from M. Spinks, NMOSE to P. Soice, Sangre de Cristo Water Co. dated June 1, 1988. Core (2003) projected CY1972-CY1987 Buckman well field pumping effects on the Rio Tesuque and Rio Pojoaque-Nambe after CY2010. Residual hydrologic effects on the Rio Grande due to CY1972-CY1987 Buckman well field pumping were calculated using the NMOSE analytical model and described in a letter from W. Fleming, NMOSE, to P. Soice, Sangre de Cristo Water Company, dated June 13, 1989.

Prior to 2004, the McAda-Wasiolek (1988) model was used to calculate impacts for 1988 through 2003 (based on pumping from CY1988 through CY2003). For the 2004 calculation (based on pumping from CY1988 through CY2004), and for subsequent annual calculations, the revised superposition version of the McAda-Wasiolek (1988) model (Core, 1996, Barroll and Keyes, 2005) was used and was run with MODFLOW96. Use of the superposition model has consolidated the number of separate model runs needed, and additionally has allowed for estimation of effects at La Cienega springs (see below). The total diversion for each of the Buckman wells was divided equally between model layers 1 and 2 and monthly pumping was simulated in the model, which is consistent with how pumping has been simulated in previous years. For this analysis, CY2022 monthly pumping from the Buckman wells was added to the well file (which contained the previously entered CY1988 though CY2021 pumping), the model was run, and the results were processed.

Table 3 summarizes the estimated streamflow depletion on the Rio Pojoaque-Nambe and Rio Tesuque from 1988 to 2025 due to Buckman well field pumping from CY1972 to CY2022. CY2022 stream depletion was calculated to be 59.844 ac-ft on the Rio Pojoaque-Nambe system, and 33.490 ac-ft on the Rio Tesuque. Starting in CY2022, maximum calendar year depletions based on Buckman well field pumping from CY1972 to CY2022 were estimated to be 60.298 ac-ft (estimated to occur in CY2024) and 33.490 ac-ft (estimated to have occurred in CY2022), in the Rio Pojoaque-Nambe system and the Rio Tesuque, respectively.

Table 4 summarizes the monthly estimated streamflow depletion on the Rio Grande above and below Otowi gage for CY2022. For CY2022, the total estimated streamflow depletion on the Rio Grande for the reach above Otowi gage is 103.41 ac-ft and for the reach below Otowi gage is 844.14 ac-ft.

Simulated flow reduction from La Cienega springs

The effect of Buckman well field pumping on La Cienega spring flow was calculated using the revised superposition version of the McAda-Wasiolek (1988) model (Core, 1996, Barroll and Keyes, 2005). The total diversion for each of the Buckman wells was divided equally between model layers 1 and 2 and monthly pumping was simulated in the model.

Table 5 summarizes the estimated depletion to La Cienega springs from 2004 to 2025 due to Buckman well field pumping from CY1988 to CY2022. CY1988 to CY2022 Buckman well field pumping is estimated to have caused a spring flow reduction at La Cienega of 3.37 ac-ft during CY2022. Based on pumping during this period, the hydrologic effect on the La Cienega Springs would reach 16.17 ac-ft starting in CY2152. It is apparent that a significant lag occurs between timing of actual Buckman pumping and the incursion of simulated maximum effect on flow at the La Cienega springs.

<u>References:</u>

Barroll, Peggy, and Keyes, Eric, 2005, *"Santa Fe Model: Resolution of Discrepancy between Superposition and Calibrated Versions of the Model"*, November 29, 2005, Office of the State Engineer Technical Memorandum to Tom Morrison, Hydrology Bureau Chief.

Compton, M.B., 1988, Untitled letter to Sangre de Cristo Water Company / V. Phillip Soice, Re: File No. RG-20516 - Buckman Well Field, dated January 14, 1988.

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Core, Andrew B., 2003, *Residual Effects of 1972-1987 pumping at Buckman well field on the Rio Tesuque and Pojoaque River*, New Mexico Office of the State Engineer Memorandum to Tom Morrison, Hydrology Bureau Chief, dated April 30, 2003.

McAda, Douglas P., and Wasiolek, Maryann, 1988, *Simulation of the Regional Geohydrology of the Tesuque Aquifer System near Santa Fe, New Mexico*, U.S. Geological Survey Water-Resources Investigations Report 87-4056.

cc: Bill Schneider, City of Santa Fe

TABLE 1. Reported Buckman Well Field pumpage data in acre-feet

Year	Well B1	Well B2	Well B3/3A	Well B4	Well B5	Well B6	Well B7	Well B8	Well B9	Well B10	Well B11	Well B12	Well B13	ANNUAL TOTAL
1988	547.85	742.16	82.49	465.34	52.44	581.03								2471.31
1989	563.10	1271.65	260.73	1049.03	146.30	1259.71								4550.52
1990	333.83	840.18	263.18	1041.18	142.50	1202.73								3823.60
1991	200.79	544.79	177.28	625.51	109.60	801.59	301.83	377.90						3139.31
1992	194.44	729.85	112.18	497.37	48.19	1215.33	993.01	961.78						4752.13
1993	692.78	735.49	137.31	647.47	140.61	1222.32	1070.03	962.28						5608.29
1994	707.94	743.24	125.12	667.62	242.34	1051.71	638.06	777.44						4953.47
1995	945.40	824.76	351.55	765.18	384.14	1009.37	861.37	778.63						5920.39
1996	935.43	707.33	616.53	631.95	437.94	972.46	592.58	761.45						5655.67
1997	924.65	815.53	467.03	453.35	139.11	1030.86	497.11	347.11						4674.75
1998	660.42	878.84	457.50	399.05	110.57	1042.80	742.71	923.63						5215.52
1999	624.75	841.62	415.54	353.42	0.00	993.70	1159.01	890.50						5278.54
2000	626.28	720.21	519.30	266.50	0.00	914.97	1169.10	864.24						5080.60
2001	716.03	489.80	436.06	354.01	0.00	1037.41	1032.51	677.52						4743.34
2002	730.45	583.98	416.69	554.62	265.43	1272.77	1145.18	868.53						5837.66
2003	508.17	540.25	337.84	386.17	234.42	1187.87	987.07	780.44	292.08	0.03	251.23	259.45	60.07	5825.09
2004	26.51	6.17	73.44	355.22	3.76	958.14	277.25	652.62	5.67	453.64	710.82	948.33	1265.27	5736.85
2005	4.99	6.39	105.14	510.34	1.04	367.40	1223.04	797.05	1.73	22.76	367.91	216.31	160.25	3784.34
2006	20.00	150.23	247.19	612.03	7.66	1076.65	237.57	814.32	54.37	656.51	824.75	88.36	407.43	5197.06
2007	398.50	8.63	183.29	565.27	0.30	408.79	703.51	704.39	25.12	222.53	136.92	31.49	404.83	3793.58
2008	75.49	0.00	115.27	782.12	60.02	37.57	851.48	866.44	0.00	262.74	556.46	42.16	84.28	3734.04
2009	29.56	0.00	143.06	704.14	157.32	951.76	530.90	367.97	0.00	233.39	330.98	345.38	7.89	3802.35
2010	81.60	0.00	437.16	474.20	128.67	1023.46	249.29	7.70	0.00	96.65	328.05	181.80	44.33	3052.91
2011	77.93	0.00	101.38	219.65	0.62	432.63	0.00	244.35	0.00	194.11	288.53	165.30	0.01	1724.51

Year	Well B1	Well B2	Well B3/3A	Well B4	Well B5	Well B6	Well B7	Well B8	Well B9	Well B10	Well B11	Well B12	Well B13	ANNUAL TOTAL
2012	213.82	0.00	95.35	149.52	48.83	316.83	0.00	75.33	0.00	20.09	0.00	130.64	0.00	1050.42
2013	631.14	4.60	218.81	403.80	0.00	514.24	31.48	255.14	4.85	184.98	288.29	159.11	194.03	2890.47
2014	54.03	0.00	81.58	33.97	29.46	134.59	53.67	20.63	70.81	0.00	29.67	40.50	25.89	574.80
2015	163.55	0.00	56.61	11.98	14.31	163.26	32.89	49.61	7.74	0.01	25.21	29.26	73.34	627.77
2016	231.96	0.00	37.16	85.96	0.01	220.13	89.25	150.83	19.23	28.69	4.41	29.59	27.60	924.82
2017	335.63	0.00	0.01	29.75	0.00	224.68	0.20	67.52	0.04	0.05	5.28	0.00	10.93	674.10
2018	361.44	0.01	16.94	44.58	0.00	160.71	0.10	55.57	14.39	33.94	40.59	14.41	88.48	831.17
2019	95.14	0.15	21.36	16.83	0.74	90.59	1.45	64.00	1.03	13.20	0.03	0.00	9.57	314.09
2020	276.18	0.00	6.64	25.38	0.02	111.77	0.09	127.51	1.82	48.51	15.68	3.20	15.80	632.60
2021	241.72	0.00	3.85	35.38	0.00	74.29	12.25	98.06	25.85	32.08	0.00	10.97	124.48	658.93
2022	458.14	0.01	0.40	38.18	0.00	7.92	0.31	196.50	11.81	45.17	89.00	0.18	127.85	975.47

Average annual pumpage, 1988 (or start-up) through 2022	391.13	348.17	203.46	407.32	83.04	687.77	483.88	487.09	26.83	127.45	214.69	134.82	156.62	3386.01
Most-recent 3-year average annual pumpage	325.35	0.00	3.63	32.98	0.01	64.66	4.22	140.69	13.16	41.92	34.89	4.78	89.38	755.67

Table 2. CY2022 monthly pumpage from the City of Santa Fe Buckman well field (in acre-feet)

WELL NUMBER	JAN 2022	FEB 2022	MAR 2022	APR 2022	MAY 2022	JUN 2022	JUL 2022	AUG 2022	SEP 2022	OCT 2022	NOV 2022	DEC 2022	TOTAL 2022
Buckman #1	0.27	13.94	167.02	25.38	0.00	0.00	0.00	100.25	33.81	75.36	2.59	39.53	458.14
Buckman #2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.01
Buckman #3A	0.00	0.39	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.40
Buckman #4	0.08	0.17	0.00	0.00	0.00	0.00	0.00	32.17	5.57	0.10	0.01	0.09	38.18
Buckman #5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Buckman #6	0.19	0.24	0.00	0.00	0.00	0.00	0.00	0.23	6.83	0.13	0.14	0.15	7.92
Buckman #7	0.11	0.11	0.00	0.00	0.00	0.00	0.00	0.09	0.00	0.00	0.00	0.00	0.31
Buckman #8	0.09	9.63	0.00	0.00	50.02	95.00	0.00	22.13	19.33	0.10	0.10	0.10	196.50
Buckman #9	0.07	0.20	0.00	0.00	0.00	0.00	0.00	11.27	0.06	0.06	0.07	0.09	11.81
Buckman #10	0.11	0.49	0.00	0.00	1.83	19.12	0.00	23.12	0.08	0.19	0.14	0.09	45.17
Buckman #11	0.00	22.03	0.00	0.00	2.85	38.43	16.76	8.94	0.00	0.00	0.00	0.00	89.00
Buckman #12	0.03	0.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.18
Buckman #13	0.10	4.79	0.00	0.00	4.93	0.00	79.02	38.56	0.12	0.11	0.10	0.11	127.85
SUM	1.05	52.14	167.02	25.38	59.63	152.55	95.78	236.77	65.80	76.06	3.15	40.15	975.47

TABLE 3. Estimated impacts from reported Buckman Well Field pumping through Calendar Year 2022

Impacts on the Rio Pojoaque-Nambe in acre-feet

Year	Residual impact (acre-feet) of 1972 - 1987 Buckman pumping (a) (Analytical Model)	Impact (acre-feet) of 1988 - 2022 reported Buckman pumping (b) [no pumping after 2022] 3D Digital Model (1988-2003) (c) Revised Superposition Model (2004-current) (d)	Total impact (acre-feet)
1988	40.432	1.060	41.492
1989	39.244	3.287	42.531
1990	37.971	5.480	43.451
1991	36.557	7.319	43.876
1992	34.928	9.960	44.888
1993	33.112	13.118	46.230
1994	31.185	15.942	47.127
1995	29.226	18.910	48.136
1996	27.296	21.873	49.169
1997	25.439	24.144	49.583
1998	23.678	26.656	50.334
1999	22.028	29.379	51.407
2000	20.491	32.254	52.745
2001	19.068	34.714	53.782
2002	17.753	37.610	55.363
2003	16.543	40.578	57.121
2004	15.429	41.763	57.192
2005	14.404	42.778** (42.781)	57.182** (57.185)
2006	13.462	44.297** (44.306)	57.759** (57.768)
2007	12.595	46.327	58.922
2008	11.797	48.426	60.223
2009	11.061	50.005	61.066
2010	10.383	51.723	62.106
2011	6.151	53.077	59.228
2012	4.693	53.850	58.543
2013	3.234	54.410	57.644
2014	1.775	55.726	57.501
2015	0.316	56.351	56.667
2016	(e)	56.944	56.944
2017		57.520	57.520
2018		58.133	58.133
2019		58.683	58.683
2020		59.154	59.154
2021		59.395	59.395
2022		59.844	59.844
2023		60.245	60.245
2024		60.298	60.298
2025		59.934	59.934

Impacts on the Rio Tesuque in acre-feet

Year	Residual impact (acre-feet) of 1972 - 1987 Buckman pumping (a) (Analytical Model)	Impact (acre-feet) of 1988 - 2022 reported Buckman pumping (b) [no pumping after 2022] 3D Digital Model (1988-2003) (c) Revised Superposition Model (2004-current) (d)	Total impact (acre-feet)
1988	21.015	0.145	21.160
1989	22.333	0.601	22.934
1990	23.391	1.166	24.557
1991	24.227	1.629	25.856
1992	24.868	2.163	27.031
1993	25.327	2.867	28.194
1994	25.615	3.497	29.112
1995	25.747	4.279	30.026
1996	25.737	4.929	30.666
1997	25.608	5.517	31.125
1998	25.378	6.154	31.532
1999	25.067	6.820	31.887
2000	24.691	7.412	32.103
2001	24.265	7.956	32.221
2002	23.8	8.659	32.459
2003	23.308	9.506	32.814
2004	22.797	10.222	33.019
2005	22.273	11.294** (11.295)	33.567** (33.568)
2006	21.743	11.967** (11.968)	33.680* (33.711)
2007	21.212	12.709	33.921
2008	20.683	13.363	34.046
2009	20.157	13.875	34.032
2010	19.639	14.526	34.165
2011	19.258	15.163	34.421
2012	18.767	15.633	34.400
2013	18.276	15.946	34.222
2014	17.785	16.474	34.259
2015	17.295	16.890	34.185
2016	16.804	17.256	34.060
2017	16.313	17.588	33.901
2018	15.822	17.974	33.796
2019	15.331	18.385	33.716
2020	14.841	18.831	33.672
2021	14.35	19.180	33.530
2022	13.859	19.631	33.490
2023	13.368	20.109	33.477
2024	12.877	20.549	33.426
2025	12.387 (f)	20.838	33.225

(a) Data from: letter to P. Soice, Sangre de Cristo Water Co., from M. Spinks, NMOSE, dated 6/1/88; and NMOSE memorandum from Andy Core to Tom Morrison dated 4/30/2003.
(b) Model-predicted streamflow depletion due to pumping from the Buckman wellfield from CY1988 through CY2022 was calculated using model cell placements for Buckman wells 5 and 8 based on 4/28/2004 GPS measurements.
(c) The McAda and Wasiolek (1988) groundwater flow model was used to calculate impacts for CY1988 through CY2003 due to simulated Buckman wellfield pumping from CY1988 through CY2003.
(d) The Revised Superposition Model (Core, 1996, Barroll and Keyes, 2005) was used to calculate impacts starting in CY2004 due to simulated Buckman wellfield pumping from CY1988 through CY2022.
(e) Residual impact of 1972 - 1987 Buckman pumping on the Rio Pojoaque-Nambe estimated at near nil by CY2017 (4/30/2003 NMOSE internal memo from Andy Core to Tom Morrison).
(f) Residual impact of 1972 - 1987 Buckman pumping on the Rio Tesuque estimated at near nil by CY2050 (4/30/2003 NMOSE internal memo from Andy Core to Tom Morrison).

** Value prior to correction of simulated pumping rates in February 2005 and February 2006 (corrected value shown in parenthesis for 2005 and 2006)

Table 4. Estimated CY2022 monthly effects on the Rio Grande due to Buckman well field pumping from CY1972 to CY2022 (in acre-feet)

	Jan-22	Feb-22	Mar-22	Apr-22	May-22	Jun-22	Jul-22	Aug-22	Sep-22	Oct-22	Nov-22	Dec-22	Total 2022
ABOVE OTOWI:													
Revised Superposition Model (pumping 1988-2021) [a]	8.81	7.94	8.77	8.48	8.77	8.49	8.77	8.78	8.50	8.79	8.52	8.80	103.41
BELOW OTOWI													
Revised Superposition Model (1988-2021) [a]	53.54	49.42	75.36	69.31	71.77	75.84	69.84	80.49	76.74	81.09	70.11	70.64	844.14

[a] Depletions due to Buckman well field pumping from 1972 to 1987 calculated using the Analytical Model are 0 ac-ft in CY2022 based on data in a from letter to P. Soice, Sangre de Cristo Water Co., from W. Fleming, NMOSE, dated 6/13/1989

TABLE 5. Estimated impacts on the springs at La Cienega due to City of Santa Fe Buckman Well Field pumping through Calendar Year 2022

Year	Impact (acre-feet) of 1988 - 2022 reported Buckman pumping (a) [no pumping after 2022] using the Revised Superposition Model (b)
2004	0.45
2005	0.66
2006	0.83
2007	0.99
2008	1.16
2009	1.32
2010	1.49
2011	1.65
2012	1.82
2013	1.97
2014	2.13
2015	2.29
2016	2.45
2017	2.60
2018	2.75
2019	2.90
2020	3.06
2021	3.21
2022	3.37
2023	3.54
2024	3.71
2025	3.86

(a) Model-predicted streamflow depletion due to pumping from the Buckman wellfield from CY1988 through CY2022 was calculated using model cell placements for Buckman wells 5 and 8 based on 4/28/2004 GPS measurements.

(b) The Revised Superposition Model (Core, 1996, Barroll and Keyes, 2005) was used to calculate impacts due to simulated Buckman wellfield pumping from CY1988 through CY2022.